Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. URL https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html.

Park, K., Choe, Y. J., and Veitch, V. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning*, 2024.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language models are unsupervised multitask learners. *OpenAI Blog*, 2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.

Rimsky, N., Gabrieli, N., Schulz, J., Tong, M., Hubinger, E., and Turner, A. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15504–15522, 2024.

Shapley, L. S. A value for n-person games. In Kuhn, H. W. and Tucker, A. W. (eds.), *Contributions to the Theory of Games II*, pp. 307–317. Princeton University Press, 1953.

Syed, A., Rager, C., and Conmy, A. Attribution patching outperforms automated circuit discovery. *arXiv preprint arXiv:2310.10348*, 2023.

Syed, A., Rager, C., and Conmy, A. Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pp. 407–416, 2024.

Todd, E., Li, M. L., Sharma, A. S., Mueller, A., Wallace, B. C., and Bau, D. Function vectors in large language models. *arXiv preprint arXiv:2310.15213*, 2023.

Turner, A. M., Thiergart, L., Udell, D., Leech, G., Mini, U., and MacDiarmid, M. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., and Shieber, S. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pp. 12388–12401, 2020.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv preprint arXiv:2211.00593*, 2022.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023.

Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

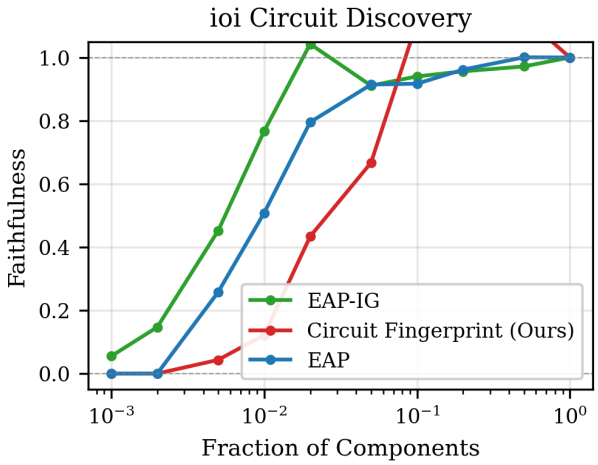
Zhang, S., Roller, S., Goyal, N., Artetxe, M., Chen, M., Chen, S., Dewan, C., Diab, M., Li, X., Lin, X. V., Mihaylov, T., Ott, M., Shleifer, S., Shuster, K., Simig, D., Koura, P. S., Sridhar, A., Wang, T., and Zettlemoyer, L. OPT: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.

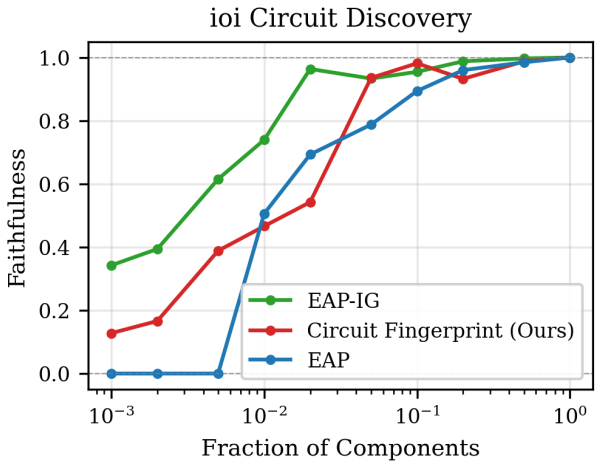
Appendix

A. Circuit Tracing Results

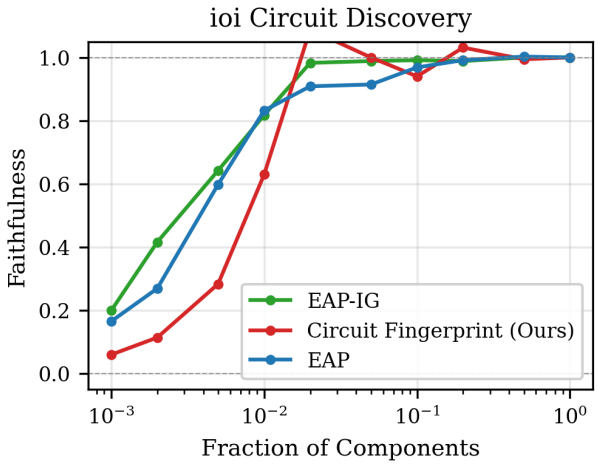
This appendix provides the complete circuit tracing results across all tasks and models discussed in Section 6.2.



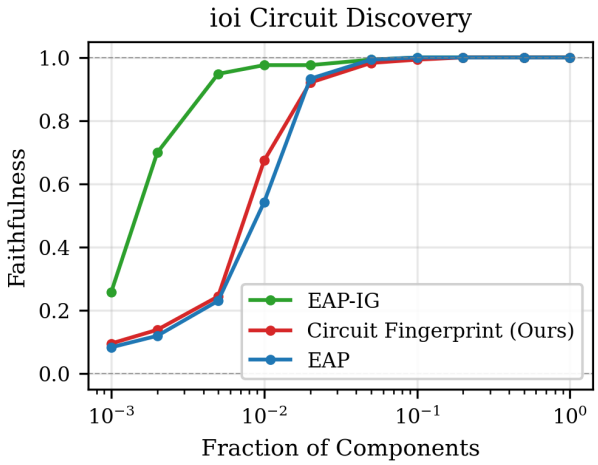
(a) GPT2-Small



(b) Qwen2.5-0.5B

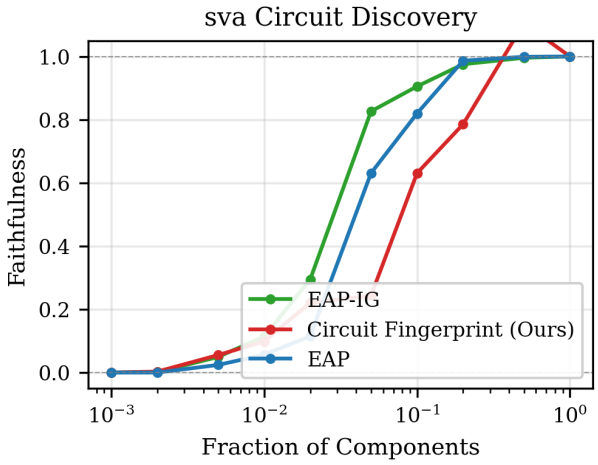


(c) Llama3.2-1B

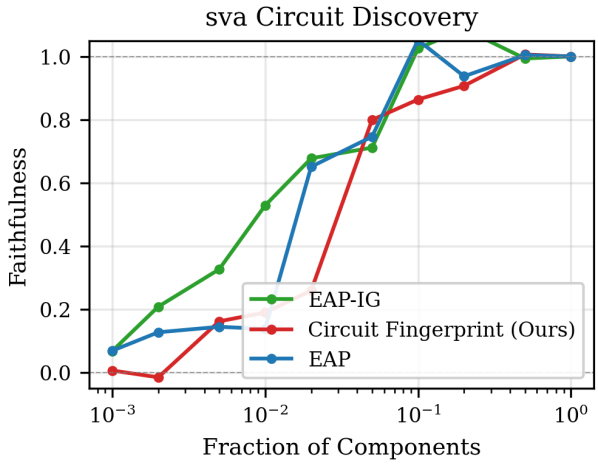


(d) OPT-1.3B

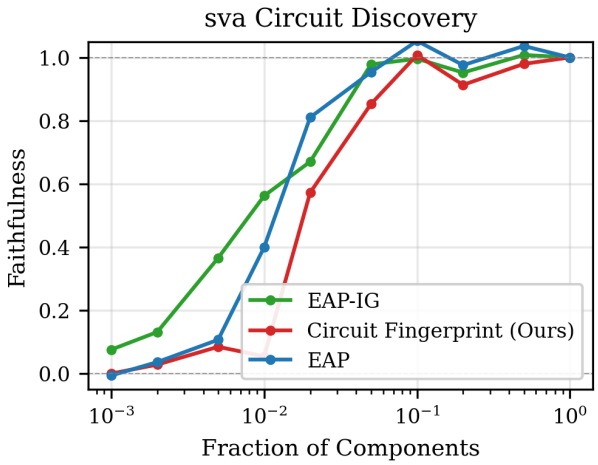
Figure 7. Edge-level circuit for IOI discovery across models.



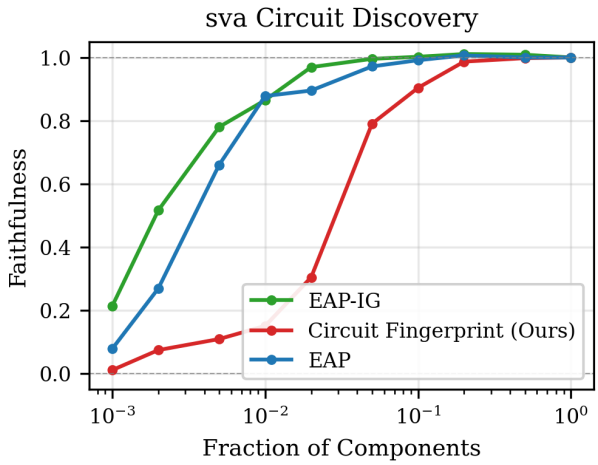
(a) GPT2-Small



(b) Qwen2.5-0.5B



(c) Llama3.2-1B



(d) OPT-1.3B

Figure 8. Edge-level circuit discovery for SVA across models.